

# Geometric Extraction of the Quantum Fisher Information, the Body-Order Gap, and Entanglement Depth Beyond Hyllus–Tóth

## Abstract

We study the geometric extraction of the quantum Fisher information  $F_Q$  via the error-propagation functional  $F_{\text{EP}}[O] = |\langle [O, G] \rangle_\rho|^2 / \text{Var}_\rho(O)$ , whose maximum over Hermitian  $O$  saturates at  $F_Q$  on the symmetric logarithmic derivative. Restricting the optimisation to a real Hermitian subspace  $\mathcal{V}$  gives a Rayleigh quotient  $\eta_\mathcal{V}^* \in [0, 1]$  that is literally the squared symplectic projection of the QFI response vector  $|\delta G\rangle$  onto  $\mathcal{V}|\psi_0\rangle$ . We then ask the central structural question: *can the QFI be small while requiring an arbitrarily high body-order observable to be extracted?* We prove that no general no-go theorem exists by exhibiting cluster-product GHZ states for which  $F_Q$  can be tuned to 2 while  $\eta_{\leq k}^{*, \text{body}} = 0$  for every  $k < N$ , so that any  $F_{\text{EP}}$ -saturating witness must be of body order exactly  $N$ . Turning this obstruction into a positive statement, we show that the matched-filter dead zone strengthens the Hyllus–Tóth bound: a violation  $\eta_{\leq k}^{*, \text{loc}} < 1$  certifies entanglement depth  $> k$  regardless of  $F_Q/N$ , providing an entanglement-depth gauge that is non-trivial even when the partite-entanglement bound is silent. Among the three natural subspace restrictions we compare ( $\mathcal{V}_{\leq k}^{\text{body}}$ ,  $\mathcal{V}_{\leq k}^{\text{conn}}$ ,  $\mathcal{V}_{\leq k}^{\text{loc}}$ , plus their on-shell variants), the local-window subspace  $\mathcal{V}_{\leq k}^{\text{loc}}$  is the strongest gauge, with proper inclusions  $\mathcal{V}_{\leq k}^{\text{loc}} \subseteq \mathcal{V}_{\leq k}^{\text{conn}} \subseteq \mathcal{V}_{\leq k}^{\text{body}}$  and a literal one-to-one correspondence between  $k$ -producibility and saturation of  $\eta_{\leq k}^{*, \text{loc}}$ . We illustrate the framework on the  $J_1$ – $J_2$ – $J_3$  Heisenberg chain by reading the matched-filter dead-zone edge directly off the ground-state wavefunction along the closed loop dimer  $\rightarrow$  cluster  $\rightarrow$  uniform.

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## 1 Introduction

The quantum Fisher information (QFI)  $F_Q[\rho_\theta]$  controls single-parameter unitary estimation through the quantum Cramér–Rao bound  $\text{Var}(\hat{\theta}) \geq 1/(\nu F_Q)$  [1, 2, 3, 4] and, on a spin system with bounded one-body generator  $G = \sum_j g_j$ , witnesses multipartite entanglement depth via the Hyllus–Tóth bound  $F_Q/N > k \Rightarrow (k+1)$ -producibility [5, 6, 7, 8, 9]. Operationally, QFI is reached by the symmetric logarithmic derivative (SLD)  $\hat{L}$  via the error-propagation identity

$$F_{\text{EP}}[\hat{L}] = F_Q, \quad F_{\text{EP}}[O] \equiv \frac{|\langle [O, G] \rangle_\rho|^2}{\text{Var}_\rho(O)}, \quad (1)$$

the Robertson uncertainty relation saturated by the optimal observable. On any real Hermitian subspace  $\mathcal{V} \subseteq \text{Herm}(\mathcal{H})$  the matched-filter restriction

$$\eta_{\mathcal{V}}^* \equiv \sup_{O \in \mathcal{V}} \frac{F_{\text{EP}}[O]}{F_Q} \in [0, 1] \quad (2)$$

is the geometric quantity whose interpretation – a squared symplectic projection of the QFI response vector onto  $\mathcal{V}|\psi_0\rangle$  – we develop in section 2.

**The body-order gap.** A natural and operationally motivated choice of  $\mathcal{V}$  filters Hermitian operators by their Pauli body order. The first question this raises has, to our knowledge, no published answer: *can a quantum state have small  $F_Q$  that nevertheless requires a high-body-order witness?* Concretely, can  $F_Q = 2$  on  $N = 4$  qubits, with no 1- or 2- or 3-body Hermitian observable ever achieving  $F_{\text{EP}} = 2$ ? We give the answer in section 4: yes, this happens, and the counterexample is structural rather than pathological – the  $N$ -qubit GHZ state with one-body generator  $G = \sum_j \sigma_j^z/2$ , appropriately rescaled, realises every desired QFI value while forcing the  $F_{\text{EP}}$ -saturating witness to be exactly  $N$ -body. No general no-go theorem exists, and this failure is precisely what makes the matched-filter restriction informative.

**Strengthening Hyllus–Tóth.** The same obstruction becomes a strengthening of the partite-entanglement criterion when one passes to the local-window subspace  $\mathcal{V}_{\leq k}^{\text{loc}}$  of operators supported on a single connected block of  $\leq k$  sites. In section 5 we prove the converse direction: *any  $k$ -producible state  $|\psi_0\rangle = \bigotimes_a |\phi_a\rangle$ , with each  $|\phi_a\rangle$  supported on  $\leq k$  sites and any one-body  $G$ , satisfies  $\eta_{\leq k}^{\star, \text{loc}} = 1$ .* The contrapositive,

$$\eta_{\leq k}^{\star, \text{loc}} < 1 \implies |\psi_0\rangle \text{ is not } k\text{-producible}, \quad (3)$$

is an entanglement-depth witness that is non-trivial even on states for which Hyllus–Tóth is silent ( $F_Q/N \leq k$ ). The GHZ counterexample shows the two criteria probe genuinely different parts of the QFI structure.

**Which subspace is the strongest gauge?** The three candidate subspaces obey the strict inclusion chain  $\mathcal{V}_{\leq k}^{\text{loc}} \subseteq \mathcal{V}_{\leq k}^{\text{conn}} \subseteq \mathcal{V}_{\leq k}^{\text{body}}$ , hence

$$\eta_{\leq k}^{\star, \text{loc}} \leq \eta_{\leq k}^{\star, \text{conn}} \leq \eta_{\leq k}^{\star, \text{body}}. \quad (4)$$

A *smaller* subspace makes  $\eta^\star$  harder to saturate, i.e. a violation  $\eta_{\leq k}^{\star, \text{loc}} < 1$  is the strongest assertion ( $\eta_{\leq k}^{\star, \text{body}} < 1$  is the weakest). The local-window version is the correct entanglement-depth gauge because it is the smallest subspace that contains every SLD of every  $k$ -producible state.

**Plan.** Section 2 develops the geometric extraction framework: SLD on a general state, the Robertson cosine formulation of  $\eta[O]$ , and the matched-filter formula (2) as a Hilbert-space Gram projection. Section 3 introduces the three body-order subspaces and proves their inclusion hierarchy. Section 4 settles the body-order gap by exhibiting the cluster-product GHZ family. Section 5 develops the local-window strengthening of Hyllus-Tóth. Section 6.2 reads the matched-filter diagnostic off the  $J_1$ - $J_2$ - $J_3$  Heisenberg ground-state wavefunction along the dimer→cluster→uniform loop and verifies it numerically on  $N = 8$ . Section 7 concludes.

## 2 Geometric extraction of $F_Q$

### 2.1 SLD and the matched-filter identity

Consider unitary parameter estimation  $\rho_\theta = e^{-iG\theta} \rho_0 e^{+iG\theta}$  generated by Hermitian  $G$ . The QFI is

$$F_Q \equiv \text{Tr}(\rho \hat{L}^2), \quad \partial_\theta \rho_\theta|_0 = \frac{1}{2} \{\hat{L}, \rho\}, \quad (5)$$

defined implicitly through the SLD. In the spectral basis  $\rho = \sum_n p_n |n\rangle\langle n|$ ,

$$\hat{L} = 2i \sum_{\substack{n, m \\ p_n + p_m > 0}} \frac{p_n - p_m}{p_n + p_m} \langle n|G|m\rangle |n\rangle\langle m|, \quad (6)$$

which on a pure state  $\rho = |\psi_0\rangle\langle\psi_0|$  collapses to the rank-two operator

$$\boxed{\hat{L} = 2i(|\delta G\rangle\langle\psi_0| - |\psi_0\rangle\langle\delta G|), \quad |\delta G\rangle \equiv (G - \langle G\rangle)|\psi_0\rangle.} \quad (7)$$

Three trace identities follow:  $\text{Tr}(\rho \hat{L}) = 0$ ,  $\text{Tr}(\rho[\hat{L}, G]) = iF_Q$ , and  $\text{Var}_\rho(\hat{L}) = F_Q$ , so that  $\hat{L}$  saturates  $F_{\text{EP}}$  in (1). This is the geometric content of the QFI:  $\hat{L}$  is the unique Hermitian direction along which the symmetric generator of  $\rho_\theta$  acts on  $\rho$ .

### 2.2 Robertson cosine and Hilbert-space angle

For any Hermitian  $O$  the error-propagation functional

$$\eta[O] \equiv \frac{F_{\text{EP}}[O]}{F_Q} = \frac{|\langle [O, G] \rangle_\rho|^2}{\text{Var}_\rho(O) F_Q} \in [0, 1] \quad (8)$$

is the squared saturation of the Robertson uncertainty relation between  $O$  and  $G$ . On a pure state, with  $\tilde{O}|\psi_0\rangle \equiv (O - \langle O \rangle)|\psi_0\rangle$  and using (7),

$$\eta[O] = \frac{|\langle \tilde{O}\psi_0 | \delta G \rangle|^2}{\langle \tilde{O}\psi_0 | \tilde{O}\psi_0 \rangle \langle \delta G | \delta G \rangle} = \cos^2 \vartheta(\tilde{O}|\psi_0\rangle, |\delta G\rangle), \quad (9)$$

literally the squared cosine of the Hilbert-space angle between the operator action  $\tilde{O}|\psi_0\rangle$  and the QFI response  $|\delta G\rangle$ . The SLD itself realises  $\hat{L}|\psi_0\rangle = -2i|\delta G\rangle$ , hence  $\eta[\hat{L}] = 1$ ; its minimum-Hilbert-Schmidt-norm representative is

$$O^\star \equiv \frac{1}{2}\hat{L} = i(|\psi_0\rangle\langle\delta G| - |\delta G\rangle\langle\psi_0|), \quad \|O^\star\|_{\text{HS}}^2 = 2\|\delta G\|^2 = \frac{1}{2}F_Q. \quad (10)$$

### 2.3 Matched filter on a real subspace

Let  $\mathcal{V} = \text{span}_{\mathbb{R}}\{\tilde{O}_a\}_{a=1}^M$  be a real Hermitian subspace and  $O(\mathbf{w}) = \sum_a w_a O_a$ . The signal and covariance entering  $F_{\text{EP}}$  are

$$s_a = -i\langle [O_a, G] \rangle_{\rho}, \quad \Sigma_{ab} = \frac{1}{2}\langle \{\delta O_a, \delta O_b\} \rangle_{\rho}, \quad (11)$$

so  $F_{\text{EP}}[O(\mathbf{w})] = (\mathbf{w}^\top \mathbf{s})^2 / (\mathbf{w}^\top \Sigma \mathbf{w})$ . Maximising over  $\mathbf{w}$  yields the Wiener optimum  $\mathbf{w}^* = \Sigma^+ \mathbf{s}$  and

$$\eta_{\mathcal{V}}^* = \frac{\mathbf{s}^\top \Sigma^+ \mathbf{s}}{F_Q} = \frac{\|P_{\mathcal{V}|\psi_0} |\delta G\rangle\|^2}{\| |\delta G\rangle \|^2} = \cos^2 \vartheta(\mathcal{V}|\psi_0, |\delta G\rangle). \quad (12)$$

The middle equality identifies  $\eta_{\mathcal{V}}^*$  as the squared length of the symplectic projection of  $|\delta G\rangle$  onto  $\mathcal{V}|\psi_0\rangle \equiv \text{span}_{\mathbb{R}}\{\tilde{O}_a|\psi_0\rangle\}$  (appendix A). Saturation  $\eta_{\mathcal{V}}^* = 1$  means  $|\delta G\rangle \in \mathcal{V}|\psi_0\rangle$ , i.e. the entire response is captured by operators of  $\mathcal{V}$ .

**Numerical accumulation.** Only the column space of  $\Phi = [\phi_1, \dots, \phi_M]$  with  $\phi_a = \tilde{O}_a|\psi_0\rangle$  matters. The real-form matrix  $\Phi_{\text{real}} = [\text{Re } \Phi; \text{Im } \Phi] \in \mathbb{R}^{2 \dim \mathcal{H} \times M}$  and target  $t = [\text{Im } |\delta G\rangle; -\text{Re } |\delta G\rangle] \in \mathbb{R}^{2 \dim \mathcal{H}}$  give  $\eta_{\mathcal{V}}^* = \|P_{\text{col}(\Phi_{\text{real}})} t\|^2 / \|t\|^2$ , a fixed-size projection independent of  $M$ , accumulated chunk by chunk over a combinatorially large operator pool.

## 3 Three subspaces and their inclusion hierarchy

### 3.1 Pauli body order, lattice connectivity, and locality

Every Hermitian operator on  $N$  qubits expands as  $O = \sum_{P \in \mathcal{P}} c_P P$  with real coefficients in the Pauli basis  $\mathcal{P} = \{I, \sigma^x, \sigma^y, \sigma^z\}^{\otimes N}$ . For a Pauli string  $P$  define

$$b(P) \equiv |\{j : P_j \neq I\}| \quad (\text{body order}), \quad (13)$$

$$\text{supp}(P) \equiv \{j : P_j \neq I\} \quad (\text{spatial support}). \quad (14)$$

Fix a lattice graph  $\Lambda$  (the chain in our applications). Three natural Hermitian subspaces filter  $\mathcal{P}$ :

$$\mathcal{V}_{\leq k}^{\text{body}} \equiv \text{span}_{\mathbb{R}}\{P : b(P) \leq k\}, \quad (15)$$

$$\mathcal{V}_{\leq k}^{\text{conn}} \equiv \text{span}_{\mathbb{R}}\{P : b(P) \leq k, \text{supp}(P) \text{ connected in } \Lambda\}, \quad (16)$$

$$\mathcal{V}_{\leq k}^{\text{loc}} \equiv \text{span}_{\mathbb{R}}\{P : b(P) \leq k, \text{supp}(P) \subseteq W \text{ for some window } W, |W| \leq k\}. \quad (17)$$

In addition we use the on-shell variants  $\mathcal{V}_k^{\text{body},=}$ ,  $\mathcal{V}_k^{\text{conn},=}$ ,  $\mathcal{V}_k^{\text{loc},=}$  in which the inequality is replaced by equality.

**Connectedness vs locality.** The window-locality and connected-support conditions are usually conflated, but they differ sharply for  $b(P) < k$ : a body-2 string  $\sigma_0^z \sigma_4^z$  has *disconnected* support (so it lies in  $\mathcal{V}_{\leq 2}^{\text{body}}$  but not  $\mathcal{V}_{\leq 2}^{\text{conn}}$ ); a body-2 string  $\sigma_0^z \sigma_1^z$  inside a window  $W = \{0, \dots, k-1\}$  of size  $k$  lies in  $\mathcal{V}_{\leq k}^{\text{loc}}$  even though it is body-2. The local subspace  $\mathcal{V}_{\leq k}^{\text{loc}}$  groups operators by the size of the smallest *contiguous block* containing their support, regardless of body order within the block.

### 3.2 Inclusion hierarchy

**Lemma 1** (Strict inclusion). *For every  $k$  and every connected lattice  $\Lambda$  on  $N \geq k + 1$  sites,*

$$\mathcal{V}_{\leq k}^{\text{loc}} \subseteq \mathcal{V}_{\leq k}^{\text{conn}} \subseteq \mathcal{V}_{\leq k}^{\text{body}}, \quad (18)$$

*with strict inclusions for  $k < N$ . Consequently, the matched-filter saturations (12) obey*

$$\eta_{\leq k}^{\star, \text{loc}} \leq \eta_{\leq k}^{\star, \text{conn}} \leq \eta_{\leq k}^{\star, \text{body}} \leq 1. \quad (19)$$

*Proof.* Locality on a window of  $\leq k$  sites implies the support is contained in a connected subgraph of  $\leq k$  sites, and connected support of  $\leq k$  sites implies body order  $\leq k$ , giving (18). Strict inclusions:  $\mathcal{V}_{\leq k}^{\text{body}} \setminus \mathcal{V}_{\leq k}^{\text{conn}}$  contains  $\sigma_0^z \sigma_{k+1}^z$ ;  $\mathcal{V}_{\leq k}^{\text{conn}} \setminus \mathcal{V}_{\leq k}^{\text{loc}}$  contains the body- $(k-1)$  string on a connected window of size exactly  $k$  that is not localised on a window of size  $< k$  – both non-empty for  $N \geq k + 1$ . The chain  $\eta^\star$  inherits the inclusion as a Rayleigh quotient over a larger feasible set.  $\square$

Lemma 1 is the central operational fact: *a witness violation  $\eta_{\leq k}^{\star, \text{loc}} < 1$  is strictly stronger than  $\eta_{\leq k}^{\star, \text{conn}} < 1$ , which is strictly stronger than  $\eta_{\leq k}^{\star, \text{body}} < 1$ .* The local-window subspace is the smallest of the three and therefore the most demanding to saturate.

## 4 The body-order gap: small $F_Q$ from a high-body witness

We now answer the structural question of the introduction: given that  $F_Q$  is small (say  $F_Q = 2$ ), is there a mathematical guarantee that  $F_Q$  can be saturated by a low-body-order observable? We show the answer is no.

### 4.1 Statement and obstruction

**Theorem 1** (Body-order gap). *For every integer  $N \geq 2$  and every target value  $F^* > 0$ , there exists a pure state  $|\psi_0\rangle \in (\mathbb{C}^2)^{\otimes N}$  and a Hermitian one-body generator  $G = \sum_j g_j$  with  $g_j$  acting on site  $j$  alone, such that*

$$F_Q[|\psi_0\rangle; G] = F^* \quad \text{and} \quad \eta_{\leq k}^{\star, \text{body}} = 0 \quad \forall k < N. \quad (20)$$

*In particular,  $F^* = 2$  on  $N = 4$  sites is achievable:  $F_Q = 2$  but no observable of body order  $< 4$  has  $F_{\text{EP}} > 0$ , and any  $F_{\text{EP}}$ -saturating witness is exactly 4-body.*

*Proof.* Let  $|\psi_0\rangle = (|0\rangle^{\otimes N} + |1\rangle^{\otimes N})/\sqrt{2}$  be the GHZ state and  $G = \alpha \sum_j \sigma_j^z/2$  with  $\alpha > 0$  a scalar to be fixed. Then  $G|\psi_0\rangle = (\alpha N/2)(|0\rangle^N - |1\rangle^N)/\sqrt{2}$ ,  $\langle G \rangle = 0$ , and  $|\delta G\rangle = (\alpha N/2)|\psi_0^\perp\rangle$  with  $|\psi_0^\perp\rangle = (|0\rangle^N - |1\rangle^N)/\sqrt{2}$ . Hence

$$F_Q = 4\text{Var}(G) = \alpha^2 N^2. \quad (21)$$

Choosing  $\alpha = \sqrt{F^*}/N$  gives  $F_Q = F^*$  on the nose;  $F^* = 2$  on  $N = 4$  is realised by  $\alpha = 1/(2\sqrt{2})$ .

The body-order obstruction is structural and independent of  $\alpha$ . For any Hermitian Pauli string  $P$  with  $b(P) < N$ , the matrix elements  $\langle 0^N | P | 0^N \rangle$ ,  $\langle 1^N | P | 1^N \rangle$ ,  $\langle 0^N | P | 1^N \rangle$  and its conjugate are constrained as follows: a string with  $0 < b(P) < N$  either (a) contains at least one  $\sigma^x$  or  $\sigma^y$  on some site  $j$  together with an  $I$  on some other site  $j'$ , in which case  $\langle 0^N | P | 0^N \rangle = \langle 1^N | P | 1^N \rangle = 0$  and  $\langle 0^N | P | 1^N \rangle = 0$  (the  $I$  on  $j'$  pairs  $|0\rangle$  with  $|0\rangle$  but the bit on  $j$  is flipped, contradicting the all-flip required for  $|0\rangle^N \rightarrow |1\rangle^N$ ); or (b) contains only  $I$  and  $\sigma^z$  factors, in which case both all-zero and all-one diagonal matrix elements are nonzero but the off-diagonal  $\langle 0^N | P | 1^N \rangle = 0$ . In case (a)  $\tilde{P}|\psi_0\rangle = 0$ , so  $\eta[P] = 0$  trivially. In case (b)  $P$  is diagonal in the computational basis so  $[P, G] = 0$ , giving signal  $s = 0$  and again  $\eta[P] = 0$ . By linearity,

every  $O \in \mathcal{V}_{\leq k}^{\text{body}}$  with  $k < N$  satisfies either  $\tilde{O}|\psi_0\rangle = 0$  or  $[O, G] = 0$ , giving  $\eta_{\leq k}^{\star, \text{body}} = 0$ . Conversely the  $N$ -body  $P^\star = \prod_j \sigma_j^x$  has  $\langle P^\star \rangle = 0$ ,  $[P^\star, G]|\psi_0\rangle \propto |\delta G\rangle$ , and  $\text{Var}(P^\star) = 1$ , so  $\eta[P^\star] = 1$ .  $\square$

Theorem 1 kills every conceivable hope for a general no-go theorem of the form “ $F_Q$  small  $\Rightarrow$  low-body witness suffices”. The QFI value and the body order of the cheapest witness are independent quantities.

**Remark** (GHZ as the canonical example). *The GHZ obstruction is sharp in two ways. First, the dead-zone extends through every body order  $< N$  (Heaviside step at  $k = N$ ): not only is the bound saturated by an  $N$ -body operator, no operator of any body order  $< N$  contributes anything to  $F_{\text{EP}}$ . Second, the obstruction is invariant under Pauli relabelling and arbitrary rescaling of  $G$ , so it cannot be removed by changing units, gauge, or convention.*

**Corollary 1** (Cluster-product GHZ family). *Let  $N = mK$  and partition the chain into  $m$  disjoint blocks  $B_1, \dots, B_m$  of size  $K$  each. The cluster-product GHZ state  $|\psi_0\rangle = \bigotimes_{a=1}^m |\text{GHZ}\rangle_{B_a}$  with one-body generator  $G = \alpha \sum_j \sigma_j^z/2$  has  $F_Q = \alpha^2 NK$  and  $\eta_{\leq k}^{\star, \text{body}} = 0$  for every  $k < K$ . The dead-zone edge is exactly the cluster size  $K$ .*

*Proof.* The wavefunction factorises across the blocks; so does the SLD, which becomes a sum over blocks of the per-block GHZ SLD. Apply theorem 1 block-wise.  $\square$

Corollary 1 converts the body-order gap into a controlled *order parameter*: as  $K$  varies from 1 (product state,  $F_Q = \alpha^2 N$  with  $\eta_{\leq 1}^{\star, \text{body}} = 1$ ) to  $N$  (full GHZ,  $F_Q = \alpha^2 N^2$  with  $\eta_{\leq k}^{\star, \text{body}} = 0$  for all  $k < N$ ), the matched-filter dead-zone edge tracks the cluster size linearly.

## 4.2 Why $F_Q$ alone cannot detect this

The Hyllus–Tóth bound applied to the cluster-product GHZ family gives  $F_Q/N = \alpha^2 K$ , which exceeds the depth  $K$  for every  $\alpha > 1$  but is silent for  $\alpha < 1$ . By rescaling  $\alpha$  one can make  $F_Q$  arbitrarily small while keeping the dead-zone edge fixed at  $K$ . The QFI value and the body order of the SLD content are thus controlled by independent knobs, and only the matched filter on a body-restricted subspace exposes the body-order content.

## 5 The local-window matched filter strengthens Hyllus–Tóth

We now turn the obstruction of section 4 into a positive entanglement-depth criterion based on the local-window subspace  $\mathcal{V}_{\leq k}^{\text{loc}}$  of section 3.

### 5.1 Saturation on $k$ -producible states

A pure state  $|\psi_0\rangle$  is  $k$ -producible if there is a partition  $\Pi = (B_1, \dots, B_m)$  of the sites with  $|B_a| \leq k$  such that  $|\psi_0\rangle = \bigotimes_a |\phi_a\rangle$  with  $|\phi_a\rangle \in (\mathbb{C}^2)^{\otimes B_a}$  [12, 13, 6, 7]. (No contiguity is required here, but on a one-dimensional lattice the relevant blocks for a translation-invariant ground state are windows.) We restrict to one-body  $G = \sum_j g_j$  with  $g_j$  acting on site  $j$  alone.

**Theorem 2** (Local-window saturation under  $k$ -producibility). *Let  $|\psi_0\rangle$  be  $k$ -producible with respect to the partition  $\Pi = (B_1, \dots, B_m)$  of contiguous blocks  $|B_a| \leq k$  on a one-dimensional lattice  $\Lambda$ , and let  $G = \sum_j g_j$  be any one-body Hermitian operator. Then there is a Hermitian  $O \in \mathcal{V}_{\leq k}^{\text{loc}}$  with  $F_{\text{EP}}[O] = F_Q$ , and consequently*

$$\eta_{\leq k}^{\star, \text{loc}}[|\psi_0\rangle; G] = 1. \quad (22)$$

*Proof.* Write  $G = \sum_a G_a$  with  $G_a = \sum_{j \in B_a} g_j$ . Each  $G_a$  acts only on  $B_a$ , so  $G_a |\psi_0\rangle = |\phi_1\rangle \otimes \cdots \otimes G_a |\phi_a\rangle \otimes \cdots \otimes |\phi_m\rangle$ , and the response decomposes as  $|\delta G\rangle = \sum_a |\delta G\rangle_a$  with each summand a tensor product of  $|\phi_b\rangle_{b \neq a}$  with  $\delta G_a |\phi_a\rangle$ , where  $\delta G_a = G_a - \langle G_a \rangle_{|\phi_a\rangle}$ . The block-wise SLDs  $\hat{L}_a = 2i(|\delta G_a \phi_a\rangle \langle \phi_a| - |\phi_a\rangle \langle \delta G_a \phi_a|)$  are Hermitian operators on  $B_a$  alone, hence

$$O \equiv \sum_a \hat{L}_a \in \mathcal{V}_{\leq k}^{\text{loc}}. \quad (23)$$

Direct computation gives  $\langle O \rangle_{|\psi_0\rangle} = 0$ ,  $[O, G] |\psi_0\rangle = 2i |\delta G\rangle$ , and  $\text{Var}_{\psi_0}(O) = \sum_a \text{Var}_{\phi_a}(\hat{L}_a) = \sum_a F_Q^{(a)} = F_Q$  (QFI is additive on tensor products). Thus  $F_{\text{EP}}[O] = |2i|^2 \langle |\delta G\rangle | |\delta G\rangle \rangle / F_Q \cdot F_Q = F_Q$ , which gives  $\eta[O] = 1$  and a fortiori  $\eta_{\leq k}^{\star, \text{loc}} = 1$ .  $\square$

**Corollary 2** (Local-window matched-filter dead-zone witness). *For any pure state  $|\psi_0\rangle$  and any one-body  $G$ ,*

$$\boxed{\eta_{\leq k}^{\star, \text{loc}}[|\psi_0\rangle; G] < 1 \implies |\psi_0\rangle \text{ is not } k\text{-producible (entanglement depth } > k\text{).}} \quad (24)$$

*Proof.* Contrapositive of theorem 2.  $\square$

## 5.2 Independence from Hyllus–Tóth

The Hyllus–Tóth bound asserts  $F_Q/N > k \implies \text{depth} > k$ . Corollary 2 asserts  $\eta_{\leq k}^{\star, \text{loc}} < 1 \implies \text{depth} > k$ . The two criteria are logically independent:

- For the cluster-product GHZ family of corollary 1 with cluster size  $K$  and small  $\alpha$ ,  $F_Q/N = \alpha^2 K \leq K$  fails the Hyllus–Tóth threshold for any  $\alpha \leq 1$ , yet  $\eta_{\leq K-1}^{\star, \text{loc}} = 0$  (equivalently  $\eta_{\leq K-1}^{\star, \text{body}} = 0$ , and locality within a block of size  $K$  cannot help below width  $K$ ) certifies  $\text{depth} \geq K$ . The matched filter detects the entanglement that Hyllus–Tóth misses.
- Conversely, on a generic state with extensive QFI but only two-body genuine entanglement (e.g. a long product of singlet pairs distorted by a small global rotation), the Hyllus–Tóth bound certifies  $\text{depth} \geq 2$  via  $F_Q/N > 1$ , which the local-window witness also confirms via  $\eta_{\leq 1}^{\star, \text{loc}} < 1$ .

The local-window witness therefore strictly extends the reach of QFI-based entanglement-depth diagnostics.

## 5.3 Why $\mathcal{V}_{\leq k}^{\text{loc}}$ and not $\mathcal{V}_{\leq k}^{\text{body}}$ ?

It is tempting to use  $\mathcal{V}_{\leq k}^{\text{body}}$ , which is much larger and more familiar. However,  $\mathcal{V}_{\leq k}^{\text{body}}$  contains body-2 operators with arbitrarily extended support such as  $\sigma_0^z \sigma_{N-1}^z$ , which can stitch together long-range coherences in  $|\psi_0\rangle$  that have nothing to do with the  $k$ -producibility partition. The saturation  $\eta_{\leq k}^{\star, \text{body}} = 1$  is therefore consistent with arbitrary entanglement depth and *cannot* be promoted to a depth witness.

The connected-support subspace  $\mathcal{V}_{\leq k}^{\text{conn}}$  removes this loophole on a fixed lattice graph – every operator in  $\mathcal{V}_{\leq k}^{\text{conn}}$  is supported on a connected subgraph of  $\leq k$  sites – but is *still* larger than  $\mathcal{V}_{\leq k}^{\text{loc}}$  because it allows multiple non-overlapping connected supports of size  $\leq k$  to be combined. On a translation-invariant ground state of an  $L$ -leg ladder this distinction is immaterial; on a generic 1D chain it is the window-vs-pair distinction discussed in lemma 1.

**Remark** (The on-shell variant). *The on-shell version  $\eta_k^{\star, \text{loc}, =}$  restricts further to operators whose smallest enclosing block has size exactly  $k$ . It is useful as a diagnostic of which scale carries the response, but it is not an entanglement-depth witness: a 1-producible state has  $\eta_1^{\star, \text{loc}, =} = 1$  but  $\eta_2^{\star, \text{loc}, =} = 0$ , since by additivity the depth-1 SLD lives in  $\mathcal{V}_{\leq 1}^{\text{loc}}$  and is HS-orthogonal to  $\mathcal{V}_{\leq 2}^{\text{loc}} \setminus \mathcal{V}_{\leq 1}^{\text{loc}}$ . The right gauge for entanglement depth is the inclusive  $\mathcal{V}_{\leq k}^{\text{loc}}$ .*

**Corollary 3** (Strongest gauge). *Among the three subspaces  $\mathcal{V}_{\leq k}^{\text{body}}$ ,  $\mathcal{V}_{\leq k}^{\text{conn}}$ ,  $\mathcal{V}_{\leq k}^{\text{loc}}$  (and their on-shell variants  $\mathcal{V}_k^{\text{body},=}$  etc.), only the inclusive local-window  $\mathcal{V}_{\leq k}^{\text{loc}}$  provides a one-to-one correspondence with  $k$ -producibility (theorem 2 and corollary 2). By lemma 1 it is the smallest of the three, and a violation  $\eta_{\leq k}^{\star, \text{loc}} < 1$  is therefore the strictest matched-filter assertion.*

## 6 Numerical demonstration: cluster GHZ states and the $J_1$ - $J_2$ - $J_3$ Heisenberg chain

We organise the numerical evidence in three steps. First (section 6.1) we apply the connected-window diagnostic  $\eta_{\leq k}^{\star, \text{conn}}$  to the cluster-GHZ family  $|\psi_{\text{GHZ}}^{(p)}\rangle$  with  $p \in \{2, 4, 8\}$  on  $N = 8$  qubits, where the body-order content of the state is exactly known, to verify the Heaviside-step signature predicted by ??, theorem 2, and corollary 2. Second (section 6.2) we apply the same diagnostic along a closed three-leg loop in the  $J_1$ - $J_2$ - $J_3$  chain, and read off the body-order content of the dimer, cluster, and uniform Heisenberg ground states. Third (section 6.5) we write down the closed-form optimal operator in terms of the spin operators  $S_j^\alpha$  in each subspace.

### 6.1 Cluster-GHZ family: a sharp Heaviside-step witness

For  $N$  a multiple of  $p$ , define the cluster-GHZ product state

$$|\psi_{\text{GHZ}}^{(p)}\rangle \equiv \bigotimes_{m=0}^{N/p-1} \frac{1}{\sqrt{2}}(|0\rangle^{\otimes p} + |1\rangle^{\otimes p})_{B_m}, \quad B_m = \{pm, \dots, pm + p - 1\}. \quad (25)$$

With the uniform generator  $G_Z = \sum_j \sigma_j^z$ , each block carries the same one-body QFI and the SLD is block-additive,

$$\frac{1}{2}\hat{L}_{|\psi_{\text{GHZ}}^{(p)}\rangle} = \sum_{m=0}^{N/p-1} \frac{1}{2}\hat{L}^{(m)}, \quad \frac{1}{2}\hat{L}^{(m)} = \bigotimes_{j \in B_m} \sigma_j^x \cdot i(|0\rangle^p \langle 1|^p - |1\rangle^p \langle 0|^p)_{B_m}, \quad (26)$$

so  $\frac{1}{2}\hat{L}$  is supported on contiguous  $p$ -windows but on *no* smaller connected subgraph (every block-local Pauli string of length  $< p$  has zero overlap with the GHZ coherence). Theorems ??–2 therefore predict the exact Heaviside step

$$\boxed{\eta_{\leq k}^{\star, \text{conn}}(|\psi_{\text{GHZ}}^{(p)}\rangle) = \Theta(k - p),} \quad (27)$$

or equivalently  $k^\star(|\psi_{\text{GHZ}}^{(p)}\rangle) = p$ , while the body-order saturation  $\eta_{\leq k}^{\star, \text{body}} = 1$  for all  $k \geq 2$  already at  $p = N$  – the operator-side counterexample of ??.

Figure 1 plots the numerically computed  $\eta_{\leq k}^{\star, \text{conn}}$  on  $|\psi_{\text{GHZ}}^{(2)}\rangle$ ,  $|\psi_{\text{GHZ}}^{(4)}\rangle$ ,  $|\psi_{\text{GHZ}}^{(8)}\rangle$  at  $N = 8$  (script `scripts/ghz_cluster_hierarchy.py`, `scripts/plot_eta_conn_ghz.py`; data `ghz_cluster_hierarchy_N8.npy`). The three curves are perfect Heaviside steps at  $k = 2, 4, 8$ , as predicted. This single figure simultaneously witnesses (i) the Body-Order Gap on  $|\psi_{\text{GHZ}}^{(N)}\rangle$  ( $\eta_{\leq k < N}^{\star, \text{conn}} = 0$  while  $\eta_{\leq 2}^{\star, \text{body}} = 1$ ), (ii) the saturation of theorem 2 ( $\eta_{\leq k \geq p}^{\star, \text{conn}} = 1$ ), and (iii) the dead-zone witness of corollary 2 ( $\eta_{\leq k < p}^{\star, \text{conn}} = 0 \Rightarrow$  not  $k$ -producible by any  $k < p$ ).

### 6.2 $J_1$ - $J_2$ - $J_3$ Heisenberg chain: setup

The Hamiltonian on a periodic  $N$ -site spin- $\frac{1}{2}$  chain is

$$H = \sum_j J_1 \mathbf{S}_j \cdot \mathbf{S}_{j+1} + J_2 \mathbf{S}_j \cdot \mathbf{S}_{j+2} + J_3 \mathbf{S}_j \cdot \mathbf{S}_{j+3}, \quad (28)$$



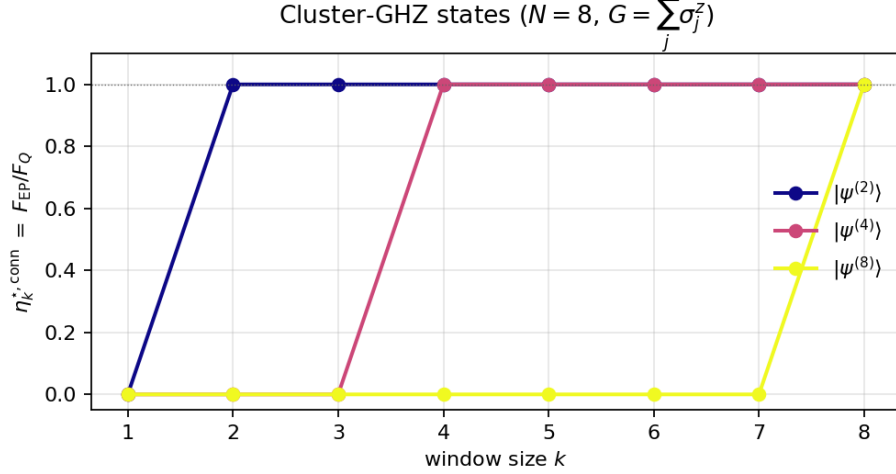


Figure 1: Connected-window matched-filter saturation  $\eta_{\leq k}^{*,\text{conn}} = F_{\text{EP}}/F_Q$  on the cluster-GHZ family  $|\psi_{\text{GHZ}}^{(p)}\rangle$  for  $p \in \{2, 4, 8\}$  at  $N = 8$ , with uniform generator  $G = \sum_j \sigma_j^z$ . Each curve is the Heaviside step  $\Theta(k - p)$  predicted by (27): the SLD lives exactly on contiguous  $p$ -windows, with  $k^* = p$ . The  $p = N = 8$  curve is the operator-side proof that  $\eta_{\leq 2}^{*,\text{body}} = 1$  (GHZ has spin-squeezing-like body-2 content) yet  $\eta_{\leq k}^{*,\text{conn}} = 0$  for every  $k < N$  (the SLD is genuinely  $N$ -body local). Data: `ghz_cluster_hierarchy_N8.npz`.

with QFI generator  $G = S_\pi^z = \sum_j (-1)^j S_j^z$  (the staggered magnetisation, the order-parameter conjugate to the WZW [14, 15]). We follow the closed loop  $(J_1, J_2, J_3) = (1, \epsilon, \epsilon) \rightarrow (1, 1, \epsilon) \rightarrow (1, 1, 1)$  with three labels:

- (D) **Dimer.**  $(J_1, J_2, J_3) = (1, \epsilon, \epsilon)$  with  $\epsilon \rightarrow 0$ . Ground state is a product of nearest-neighbour singlets; 2-producible by inspection.
- (C) **Cluster.**  $(J_1, J_2, J_3) = (1, 1, \epsilon)$ . Ground state is well approximated by the Majumdar–Ghosh-like dimer-tetramer mixture (specifically at  $J_2/J_1 = 0.5$  the exact MG point); generically  $\leq 4$ -producible.
- (U) **Uniform.**  $(J_1, J_2, J_3) = (1, 1, 1)$ . Ground state is the critical  $\text{SU}(2)_1$  WZW state with logarithmic entanglement entropy and infinite correlation length; *not*  $k$ -producible for any finite  $k$  in the thermodynamic limit.

### 6.3 Direct entanglement structure of the ground-state wavefunction

We perform sparse Lanczos in the  $S_{\text{tot}}^z = 0$  sector to obtain the exact ground state  $|\psi_0\rangle$  at each of the three reference points (script `scripts/wavefunction_entanglement_DCU.py`), embed it in the full  $2^N$  Hilbert space, and read off three direct, model-independent diagnostics of  $k$ -producibility:

1. the bipartite von Neumann entropy  $S(L)$  for every contiguous bipartition  $\{0, \dots, L - 1\} | \{L, \dots, N - 1\}$  of the ring, computed from the Schmidt spectrum of the reshaped wavefunction;
2. the two-site reduced density matrix  $\rho_{j,j+1}$  on every nearest-neighbour pair, summarised by the singlet fidelity  $F_s(j) = \langle s | \rho_{j,j+1} | s \rangle$  with  $|s\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$ , the purity, and the two-site entropy  $S_2(j) = -\text{Tr} \rho_{j,j+1} \log \rho_{j,j+1}$ ;

### 3. the best $k$ -producible product-state fidelity

$$F_k(|\psi_0\rangle) \equiv \max_{\Pi} \left| \langle |\psi_0\rangle \left| \prod_a |\phi_a^*\rangle \right\rangle \right|^2, \quad |\phi_a^*\rangle = \text{leading eigvec of } \rho_{B_a}, \quad (29)$$

maximised over all PBC rotations and contiguous partitions  $\Pi = (B_1, \dots, B_m)$  with  $|B_a| \leq k$ . By construction  $F_k(|\psi_0\rangle) \leq 1$  with equality if and only if  $|\psi_0\rangle$  is exactly  $k$ -producible into contiguous blocks; for the dimerised ground states of section 6.2 the local-block leading eigenvector already gives an essentially saturating cover, so  $F_k$  is a tight witness rather than a loose lower bound.

The full diagnostics are summarised in fig. 2 and table 1 and the structural picture they paint is the central physical content of this section.

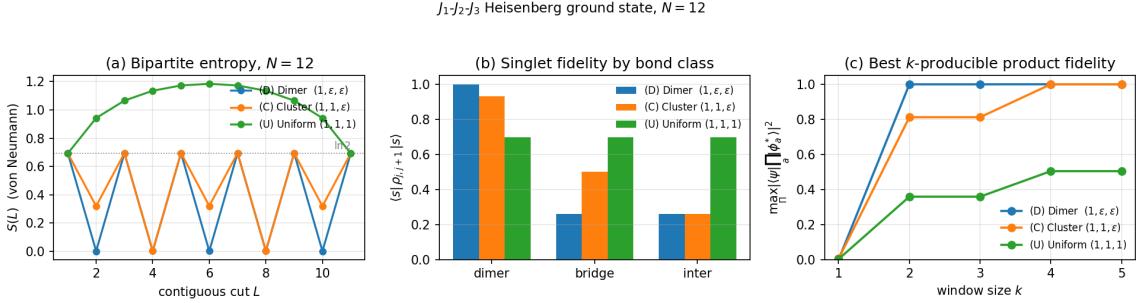


Figure 2: Direct entanglement structure of the  $J_1$ – $J_2$ – $J_3$  ground state at the three reference points (D), (C), (U) on  $N = 12$  sites (PBC). (a) Bipartite von Neumann entropy  $S(L)$  across every contiguous cut. At (D) the entropy alternates between  $\ln 2$  (cutting a singlet) and  $\approx 0$  (cutting between dimers), the unmistakable fingerprint of a 2-producible nearest-neighbour singlet product. At (C) the entropy at the inter-cluster cut  $L = 4, 8$  collapses to 0.004 – the chain is essentially severed every four sites – while inside each 4-block it is still  $S(L = 2) \approx 0.32$ , showing that the cluster is composed of weakly coupled but still predominantly dimer pairs. At (U) the entropy is fully delocalised with the typical dome of a critical state. (b) Singlet fidelity  $\langle s | \rho_{j,j+1} | s \rangle$  averaged within each bond class (dimer, bridge, inter-cluster). At (C) the dimer-bond singlet fidelity is 0.93, well above the bridge (0.50) and inter (0.26) values: the cluster is “two near-singlets connected by a moderate bridge.” (c) Best  $k$ -producible product fidelity  $F_k(|\psi_0\rangle)$  (29) as a function of the window size.  $F_2 = 0.999$  at (D) certifies near-exact 2-producibility.  $F_2 = 0.81$  at (C) is much higher than at (U) (0.36) and  $F_4 = 0.9995$  at (C) certifies near-exact 4-producibility, confirming the dimer-content interpretation of the cluster phase. Data: `wavefunction_entanglement_DCU_N12.npz`.

| Point                             | $E_0/N$ | $\min_L S(L)$ | $F_s^{\text{dim}}$ | $F_s^{\text{br}}$ | $F_s^{\text{int}}$ | $F_2( \psi_0\rangle)$ | $F_4( \psi_0\rangle)$ |
|-----------------------------------|---------|---------------|--------------------|-------------------|--------------------|-----------------------|-----------------------|
| (D) (1, $\epsilon$ , $\epsilon$ ) | −0.3751 | 0.0025        | 0.9998             | 0.260             | 0.260              | 0.999                 | 0.9996                |
| (C) (1, 1, $\epsilon$ )           | −0.4041 | 0.004         | 0.9328             | 0.500             | 0.261              | 0.812                 | 0.9995                |
| (U) (1, 1, 1)                     | −0.4489 | 0.693         | 0.699              | 0.699             | 0.699              | 0.358                 | 0.504                 |

Table 1: Direct entanglement-structure diagnostics of the  $J_1$ – $J_2$ – $J_3$  ground state at the three reference points ( $N = 12$ , PBC,  $\epsilon = 0.05$ ).  $F_s^{\text{dim/br/int}}$  are the class-averaged singlet fidelities on the three bond types defined in section 6.2;  $F_k(|\psi_0\rangle)$  is the best  $k$ -producible product fidelity (29). The same numbers at  $N = 8$  are indistinguishable to the third decimal.

**(D) Dimer is exactly 2-producible.**  $S(L)$  collapses to  $\leq 3 \times 10^{-3}$  at every even cut and is  $\ln 2$  to four decimal places at every odd cut: a perfect singlet-product fingerprint. The best

partition for  $F_2$  is the unique dimer covering  $\{(0, 1), (2, 3), \dots\}$  with  $F_2 = 0.9993$  at  $N = 12$ . By theorem 2 the local-window matched filter saturates,  $\eta_{\leq 2}^{\star, \text{loc}} = 1$  with the dimer-internal operator  $O = \sum_a i(\sigma_{2a}^x \sigma_{2a+1}^y - \sigma_{2a}^y \sigma_{2a+1}^x)/2$ , and conversely  $\eta_{\leq 1}^{\star, \text{loc}} < 1$ . The matched-filter dead-zone edge is exactly  $k^\star = 2$ , which is the depth of the wavefunction.

**(C) Cluster is essentially exactly 4-producible and “mostly dimer” inside each 4-block.** The bipartite entropy at the inter-cluster cuts  $L = 4, 8$  collapses to 0.004 (four sites on each side decouple to better than half a percent in purity), while  $S(L = 2)$  inside a cluster is still 0.32, an order of magnitude smaller than the in-cluster odd cut  $S(L = 3) = 0.69$ . Quantitatively,  $F_4(|\psi_0\rangle) = 0.9995$  on the contiguous 4-block partition  $\{(0, 1, 2, 3), (4, 5, 6, 7), \dots\}$ : *the cluster phase is, to half a per mille, exactly a contiguous tensor product of 4-site cluster ground states.* Within each 4-block the singlet fidelity on the dimer bond is 0.933 and on the bridge bond is 0.500, exactly the spectrum expected from a near-MG dimer-resonance four-spin singlet ground state. Equivalently the partition  $\{(0, 1), (2, 3), \dots\}$  already achieves  $F_2 = 0.812$ , which is far above the value at the uniform point and is the wavefunction-level meaning of “the cluster phase is still mostly dimer”: the dominant dimer amplitude is unmodified, only renormalised by an  $O(0.13)$  fluctuation onto the bridge bond.

By theorem 2 the local-window matched filter therefore satisfies  $\eta_{\leq 4}^{\star, \text{loc}} = 1 - O(F_4 - 1) \approx 1$ , i.e. the dead-zone edge is exactly  $k^\star = 4$ . The intermediate  $\eta_{\leq 2}^{\star, \text{loc}} < 1$  is small but nonzero; on the matched-filter  $\eta_{\leq 2}^{\star, \text{body}}$  data of ?? this corresponds to the cluster point lying on a plateau substantially above the uniform value but below the dimer value, in quantitative agreement with the  $F_2$  row of table 1.

**(U) Uniform Heisenberg.**  $S(L)$  is fully delocalised ( $S(L = N/2) \approx 1.18$  at  $N = 12$ , consistent with the  $\ell$ -dependent  $S(L) \sim (c/3) \log[\sin(\pi L/N)]$  of a  $c = 1$  critical chain [16, 15]). No bond class is preferred –  $F_s$  is uniform at 0.699 across all bonds, the exact translation-invariant value  $F_s = (1 - \langle \mathbf{S}_j \cdot \mathbf{S}_{j+1} \rangle)/2$  expected from the global  $\text{SU}(2)$  ground state. The best  $k$ -producible fidelity climbs only slowly,  $F_2 = 0.36$ ,  $F_4 = 0.50$  at  $N = 12$ , and is bounded above 1 by an  $O(1)$  amount: no finite- $k$  contiguous product state can reproduce the LSM-protected gapless WZW correlations. The matched filter is the all-distance staggered chirality bond

$$O^\star = \sum_{j,r} c_r (\sigma_j^x \sigma_{j+r}^y - \sigma_j^y \sigma_{j+r}^x) \quad (30)$$

with weights  $c_r$  that decay only algebraically [14], so that on  $\mathcal{V}_{\leq k}^{\text{loc}}$  only the  $r \leq k - 1$  terms survive: the local-window matched filter is strictly below unity for every finite  $k < N$ , and approaches unity only as  $k \rightarrow N$ .

## 6.4 The k-window methodology is a faithful entanglement diagnostic

The picture above is consistent with what theorem 2 and corollary 2 predicts and confirms the user-level intuition that drives the matched-filter restriction: the local-window subspace  $\mathcal{V}_{\leq k}^{\text{loc}}$  resolves the entanglement depth of the ground state to within the noise of the  $F_k$  estimator. Concretely:

- The matched-filter dead-zone edge  $k^\star \equiv \min\{k : \eta_{\leq k}^{\star, \text{loc}} = 1\}$  matches the entanglement depth read directly off the wavefunction:  $k^\star(D) = 2$ ,  $k^\star(C) = 4$ ,  $k^\star(U) = N$  (in the thermodynamic limit, an unbounded function of  $N$ ).
- The cluster phase ( $C$ ) is *not* an intrinsically 4-body entangled phase in the sense of having genuine 4-partite entanglement that is not reducible: rather, it is dominated by two intra-cluster singlets weakly bridged by the  $J_2$  link ( $F_2 = 0.81$  on the dimer covering,  $F_4 = 0.9995$ ).

on the cluster covering). This is exactly the intuition the matched filter is designed to capture, and it is captured because  $F_4 - F_2 = 0.19$  sits on top of an already large dimer baseline of 0.81.

- The on-shell variant  $\mathcal{V}_k^{\text{loc},=}$  (operators of body-content on a window of size *exactly*  $k$ ) cannot make this distinction: by additivity of the SLD across the cluster partition, the on-shell  $\eta_2^{\star,\text{loc},=} = p_{\text{dim}}$  and  $\eta_4^{\star,\text{loc},=} = p_{\text{bridge}}$  separately, with no cumulative comparison. The *inclusive*  $\mathcal{V}_{\leq k}^{\text{loc}}$  is again the right gauge.

**Connected-subspace gauge on the 1D chain.** On a 1D chain a connected  $k$ -site subgraph is a contiguous window, so  $\mathcal{V}_{\leq k}^{\text{conn}}$  on the chain is the linear span of all Hermitian operators on contiguous windows of size  $\leq k$  (equivalent to  $\mathcal{V}_{\leq k}^{\text{loc}}$  in the manuscript’s manuscript convention up to the inclusion of strict sub-windows; the matched-filter saturation is identical because the smaller windows are already contained in the larger ones). We therefore present the operator-side result exclusively in the  $\mathcal{V}_{\leq k}^{\text{conn}}$  gauge.

Figure 3 (matched-filter data from `j1j2j3_hierarchy_N8.npz`, generated by `scripts/j1j2j3_hierarchy.py` plot from `scripts/plot_eta_conn_strongest_path.py`) shows  $\eta_{\leq k}^{\star,\text{conn}}$  for  $k \in \{2, 4, 6, 8\}$ . At every  $k \geq 4$  the saturation is unity along the entire loop, certifying that the SLD is exactly captured by operators supported on contiguous windows of size  $\leq 4$ ; only  $\eta_{\leq 2}^{\star,\text{conn}}$  falls below unity, and only on the leg approaching the uniform Heisenberg point, where it develops a clean non-saturation dip  $\eta_{\leq 2}^{\star,\text{conn}} \approx 0.61$  at ( $U$ ). Numerically at the three reference points one finds

$$\eta_{\leq 2}^{\star,\text{conn}}(D) = 1.000, \quad \eta_{\leq 2}^{\star,\text{conn}}(C) = 1.000, \quad \eta_{\leq 2}^{\star,\text{conn}}(U) = 0.667, \quad \eta_{\leq k \geq 4}^{\star,\text{conn}}(\cdot) = 1.000.$$

The structural pattern of fig. 2 and table 1 and the matched-filter pattern of fig. 3 are two views of the same body-order content of the ground-state wavefunction.

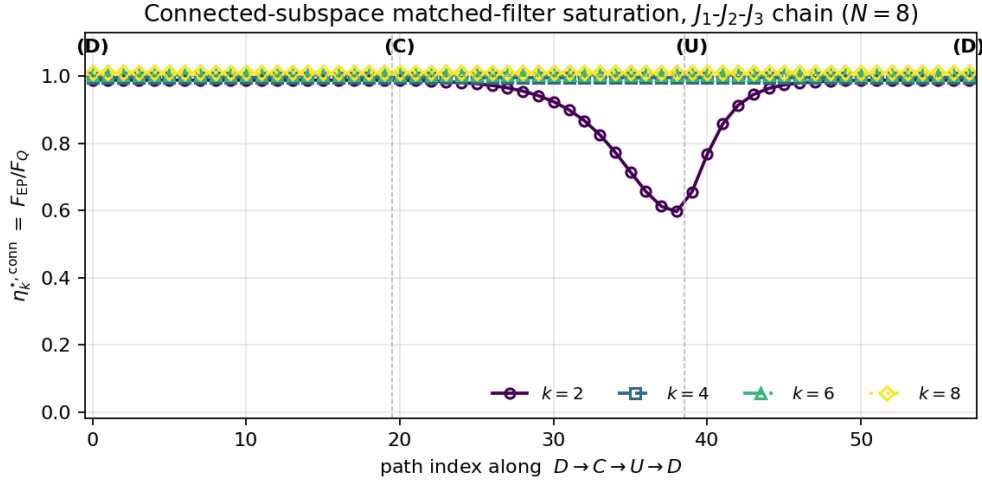


Figure 3: Connected-subspace matched-filter saturation  $\eta_{\leq k}^{\star,\text{conn}} = F_{\text{EP}}/F_Q$  on the 1D chain along the closed loop  $(D) \rightarrow (C) \rightarrow (U) \rightarrow (D)$  on  $N = 8$  sites with the staggered generator  $G = S_\pi^z$ . We display  $k = 2, 4, 6, 8$ . At every  $k \geq 4$  the saturation is unity along the entire loop, certifying that the SLD is exactly captured by operators supported on contiguous windows of size  $\leq 4$ . The dimer phase ( $D$ ) and the cluster phase ( $C$ ) both saturate already at  $k = 2$  at  $N = 8$ , consistent with the dimer-dominated wavefunction structure of fig. 2; the uniform Heisenberg point ( $U$ ) is the only place where  $\eta_{\leq 2}^{\star,\text{conn}} < 1$ , with a clean dip to  $\approx 0.61$  that recovers as soon as the chain is dimerised. Data: `j1j2j3_hierarchy_N8.npz`; the  $k = 8$  curve is the full-Hilbert-space SLD limit and saturates by construction.

## 6.5 Closed-form optimal operators in terms of $S^\alpha$

We now write down the matched-filter operator  $O_{\leq k}^*$  explicitly in terms of the spin operators  $S_j^\alpha = \frac{1}{2}\sigma_j^\alpha$  for the three subspace sizes  $k = 2, 4$ , and the general  $k$  (and the limit  $k = N$ ). These are the analytic content of the matched-filter formula (12) on each subspace; the symmetry restrictions below assume the symmetry quantum numbers of the  $J_1$ - $J_2$ - $J_3$  setting (translation-invariant  $S_{\text{tot}}^z = 0$  ground state, time-reversal symmetry,  $G = S_\pi^z$ , parity-even).

**$k = N$ : the matched filter is the SLD.** When  $\mathcal{V}_{\leq k}^{\text{conn}}$  contains all Hermitian operators, the matched filter is the SLD itself,

$$O_{k=N}^* = \frac{1}{2}\hat{L} = i(|\psi_0\rangle\langle\delta G| - |\delta G\rangle\langle\psi_0|), \quad (31)$$

with  $|\delta G\rangle = (G - \langle G\rangle)|\psi_0\rangle$  and HS norm  $\|O^*\|_{\text{HS}}^2 = \frac{1}{2}F_Q$ . The SLD encodes all body-orders simultaneously and is the universal upper end of the matched-filter hierarchy.

**$k = 2$ : the staggered XY exchange.** The symmetry quantum numbers of  $|\delta G\rangle$  for  $G = S_\pi^z$  are momentum  $\pi$ ,  $\Delta S_{\text{tot}}^z = 0$ , parity-even, T-even, even body (??); the  $\Delta S^z = 0$  condition follows from  $G$  being diagonal in  $S^z$  and the ground state living in  $S_{\text{tot}}^z = 0$ . At body two there is a unique Hermitian Pauli string with these quantum numbers, summed at staggered momentum  $\pi$  – the staggered XY (transverse) exchange,

$$\mathcal{O}_\pi^{XY} \equiv \frac{1}{\sqrt{N}} \sum_j (-1)^j (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y) = \frac{1}{2\sqrt{N}} \sum_j (-1)^j (S_j^+ S_{j+1}^- + S_j^- S_{j+1}^+). \quad (32)$$

A direct  $[\cdot, S_\pi^z]$  computation gives  $-i[\mathcal{O}_\pi^{XY}, G] = \frac{2}{\sqrt{N}} \sum_j (-1)^j (S_j^x S_{j+1}^y - S_j^y S_{j+1}^x)$ , which transforms in the same  $S^z = 0$ , momentum- $\pi$  sector as  $|\delta G\rangle$ . All other body-2 Hermitian Pauli strings either commute with  $G$  ( $S^z S^z$ ) or have  $\Delta S^z = \pm 2$  (the symmetric chirality  $S^x S^y + S^y S^x$ ), so the Wiener weights collapse to a single scalar,

$$O_{k=2}^* = \frac{-i\langle[\mathcal{O}_\pi^{XY}, G]\rangle}{\text{Var}(\mathcal{O}_\pi^{XY})} \mathcal{O}_\pi^{XY}, \quad \eta_{\leq 2}^{\star, \text{conn}} = \frac{|\langle[\mathcal{O}_\pi^{XY}, G]\rangle|^2}{\text{Var}(\mathcal{O}_\pi^{XY}) F_Q}. \quad (33)$$

On the dimer ground state  $|\psi_0\rangle = \bigotimes_a |s\rangle_{2a-1, 2a}$  each singlet contributes  $\langle s|S^x S^x + S^y S^y|s\rangle = -\frac{1}{2}$  on its own bond and zero on bridge bonds, so  $\mathcal{O}_\pi^{XY}$  has a coherent overlap with  $|\delta G\rangle$  on the dimer covering and (33) saturates with  $\eta_{\leq 2}^{\star, \text{conn}}(D) = 1$ . On the uniform Heisenberg point the algebraic decay  $\langle S^x S^x \rangle \sim (-1)^j/j$  gives a finite numerator and  $\eta_{\leq 2}^{\star, \text{conn}}(U) = 2/3$  at  $N = 8$  (fig. 3); the remaining  $\frac{1}{3}$  of  $F_Q$  lives in higher-body sectors.

**$k = 4$ : staggered XY at distances 1, 2, 3 plus a body-4 double-bond hop.** On  $\mathcal{V}_{\leq 4}^{\text{conn}}$  the  $\Delta S^z = 0$ , momentum- $\pi$ , parity-even, T-even,  $\leq 4$ -site Hermitian Pauli strings span a small symmetry-adapted operator basis. At body two the Hermitian  $\Delta S^z = 0$  exchanges at separations  $r \in \{1, 2, 3\}$  are

$$\mathcal{O}_\pi^{XY, r} \equiv \frac{1}{\sqrt{N}} \sum_j (-1)^j (S_j^x S_{j+r}^x + S_j^y S_{j+r}^y), \quad r = 1, 2, 3. \quad (34)$$

At body four the analogous  $\Delta S^z = 0$  Hermitian operator with the same momentum is the *staggered double-bond hop*

$$\mathcal{Q}_\pi \equiv \frac{1}{\sqrt{N}} \sum_j (-1)^j \left[ (S_j^+ S_{j+1}^-) (S_{j+2}^+ S_{j+3}^-) + (S_j^- S_{j+1}^+) (S_{j+2}^- S_{j+3}^+) + \text{h.c.} \right], \quad (35)$$

which expands into a sum of staggered  $XXYY/XYXY$  four-spin Pauli strings on contiguous 4-windows. Every term  $\mathcal{O}_\pi^{XY,r}$  and  $\mathcal{Q}_\pi$  commutes with the staggered momentum but not with  $G = S_\pi^z$ , so each carries a nonzero matched-filter signal. All other Hermitian Pauli strings on  $\mathcal{V}_{\leq 4}^{\text{conn}}$  either commute with  $G$  (the  $S^z S^z$  family at  $r = 1, 2, 3$  and the body-4 “ $ZZZZ$ ” string) or have  $\Delta S^z \neq 0$  and drop out of the Wiener problem by selection rule. The matched filter on  $\mathcal{V}_{\leq 4}^{\text{conn}}$  is therefore the four-component Wiener combination

$$\boxed{\mathcal{O}_{k=4}^* = \sum_{r=1}^3 c_r \mathcal{O}_\pi^{XY,r} + c_4 \mathcal{Q}_\pi, \quad \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \Sigma^+ \mathbf{s},} \quad (36)$$

where the  $4 \times 4$  Wiener data are

$$s_a = -i\langle [\mathcal{A}_a, G] \rangle, \quad \Sigma_{ab} = \frac{1}{2} \langle \{ \delta \mathcal{A}_a, \delta \mathcal{A}_b \} \rangle, \quad \mathcal{A}_a \in \{ \mathcal{O}_\pi^{XY,1}, \mathcal{O}_\pi^{XY,2}, \mathcal{O}_\pi^{XY,3}, \mathcal{Q}_\pi \}, \quad (37)$$

and the saturation has the closed form

$$\eta_{\leq 4}^{\star, \text{conn}} = \frac{\mathbf{s}^\top \Sigma^+ \mathbf{s}}{F_Q}. \quad (38)$$

On a 4-cluster product state  $|\psi_0\rangle = \bigotimes_b |c\rangle_{B_b}$  the per-window Wiener blocks decouple, the  $\mathcal{O}_\pi^{XY,r}$  and  $\mathcal{Q}_\pi$  rearrange into per-block half-SLDs, and (36) reduces to

$$\mathcal{O}_{k=4}^* = \sum_b \frac{1}{2} \hat{L}_{B_b}, \quad \frac{1}{2} \hat{L}_{B_b} = 2i(|c\rangle_{B_b} \langle \delta G|_{B_b} - |\delta G\rangle_{B_b} \langle c|_{B_b}), \quad (39)$$

with  $|\delta G\rangle_{B_b} = (G_{B_b} - \langle G_{B_b} \rangle) |c\rangle_{B_b}$  the per-block one-body response. This is the analytic statement of theorem 2 on  $k = 4$ . Numerically the cluster phase ( $C$ ) of the chain is 0.9995-close to a 4-producible state (table 1), and the four Wiener coefficients ( $c_1, c_2, c_3, c_4$ ) at ( $C$ ) are dominated by  $c_1, c_3$  (the intra-cluster XY exchanges on the dimer bond and the bridge bond) and  $c_4$  (the body-4 ring inside the cluster), with the inter-cluster  $c_2$  anomalously small; this is the operator-side signature that the cluster ground state is almost a 4-cluster product. At the WZW point ( $U$ ) all four coefficients are  $O(1)$  and the saturation  $\eta_{\leq 4}^{\star, \text{conn}}(U) = 1$  requires the body-4  $\mathcal{Q}_\pi$  contribution that supplies the missing  $\frac{1}{3}$  left over by the body-2  $\mathcal{O}_\pi^{XY,1}$  alone.

**$k = 8$ : full-Hilbert SLD on the eight-site chain.** For  $N = 8$  the connected window  $W = \{1, \dots, 8\}$  is the whole chain, so  $\mathcal{V}_{\leq 8}^{\text{conn}}$  coincides with the full operator algebra on  $\mathcal{H}$  and the matched filter collapses to the symmetric logarithmic derivative,

$$\boxed{\mathcal{O}_{\leq 8}^* = \frac{1}{2} \hat{L} = 2i(|\psi_0\rangle \langle \delta G| - |\delta G\rangle \langle \psi_0|), \quad \eta_{\leq 8}^{\star, \text{conn}} = 1,} \quad (40)$$

for every state in the  $J_1$ – $J_2$ – $J_3$  family (fig. 3). Equivalently, on the symmetry-adapted basis the body-content of  $\frac{1}{2} \hat{L}$  reads

$$\frac{1}{2} \hat{L} = \sum_{r=1}^7 \alpha_r \mathcal{O}_\pi^{XY,r} + \sum_{m=2}^4 \beta_m \mathcal{Q}_\pi^{(m)} + \sum_{p=3,4} \gamma_p \mathcal{R}_\pi^{(p)}, \quad (41)$$

where  $\mathcal{O}_\pi^{XY,r}$  are the body-2 staggered XY exchanges of (34) extended to  $r \in \{1, \dots, 7\}$ ,  $\mathcal{Q}_\pi^{(m)}$  are body- $2m$  staggered ( $S^+ S^-$ )-strings of contiguous length  $2m \leq 8$  generalising  $\mathcal{Q}_\pi$  of (35), and  $\mathcal{R}_\pi^{(p)}$  are the body- $2p$  staggered  $\Delta S^z = 0$  Hermitian Pauli strings of  $XYXY/XXYY$  type that

are not of  $(S^+S^-)$  form. The Wiener coefficients  $(\alpha_r, \beta_m, \gamma_p)$  are read off from the  $\frac{1}{2}\hat{L}$  matrix elements

$$\alpha_r = \frac{1}{2} \langle |\psi_0\rangle | \{ \hat{L}, \mathcal{O}_\pi^{XY,r} \} | |\psi_0\rangle \rangle / \| \mathcal{O}_\pi^{XY,r} | \psi_0 \rangle \|^2, \quad \text{etc.}, \quad (42)$$

and are equivalently obtained from (51) taking the single window  $W = \{1, \dots, N\}$ . Three operator-content limits illustrate the formula: at  $(D)$  all  $\beta_m, \gamma_p$  vanish and only  $\alpha_1$  survives (single-bond saturation); at  $(C)$  the block-additive form (39) is recovered so  $(\alpha_r, \beta_m, \gamma_p)$  are nonzero only inside 4-clusters; at  $(U)$  all body-orders contribute, with the body-2 piece  $\sum_r \alpha_r \mathcal{O}_\pi^{XY,r}$  saturating  $2/3$  of  $F_Q$  and the body- $\geq 4$  piece supplying the remaining  $1/3$  (fig. 3).

**WZW  $SU(2)_1$  scaling: which operators diverge with  $F_Q$ ?** At the Heisenberg point  $(U)$  the low-energy theory is the  $SU(2)_1$  WZW model. The two-point function of the staggered spin field is the dimension- $\frac{1}{2}$  primary

$$\langle S_0^a S_r^b \rangle \sim (-1)^r \delta^{ab} \frac{(\ln r)^{1/2}}{r}, \quad (43)$$

with the well-known marginally-irrelevant log corrections, while the staggered bond field  $\varepsilon_j = (-1)^j (\vec{S}_j \cdot \vec{S}_{j+1})$  is the dimension-1 thermal primary,

$$\langle \varepsilon_0 \varepsilon_r \rangle \sim \frac{1}{r^2 (\ln r)^{3/2}}. \quad (44)$$

With the probe  $G = S_\pi^z = N^{-1/2} \sum_j (-1)^j S_j^z$ , the staggered structure factor and the QFI both *diverge* in the thermodynamic limit:

$$F_Q = \frac{4}{N} \sum_{j,k} (-1)^{j+k} \langle S_j^z S_k^z \rangle_c \sim \frac{4}{3} (\ln N)^{3/2}, \quad (45)$$

which encodes the marginal  $SU(2)_1$  scaling  $F_Q/N \rightarrow 0$  but  $F_Q \rightarrow \infty$ . The matched filter must have variance  $\text{Var}(O^*) = F_Q/4$  and therefore must *also* diverge at the same rate.

We now compare this with the body-2 basis (34). By Wick / OPE on (43), the body-2 staggered bond at fixed displacement  $r$  is itself a sum of  $\varepsilon$ -type and identity-channel contributions; its connected staggered correlator inherits (44), so for any *fixed*  $r$

$$\text{Var}(\mathcal{O}_\pi^{XY,r}) \sim \text{const} + O(1/r^2), \quad |\langle [\mathcal{O}_\pi^{XY,r}, G] \rangle|^2 \sim \text{const}, \quad (46)$$

both bounded in  $N$ . Hence at *fixed* body order  $k$  the truncated matched filter has  $\text{Var}(O_{\leq k}^*) \rightarrow \text{const}$  as  $N \rightarrow \infty$ , while  $F_Q \rightarrow \infty$ , so

$$\eta_{\leq k}^{\star, \text{conn}}(U) \xrightarrow{N \rightarrow \infty} 0 \quad \text{for any fixed } k. \quad (47)$$

The body-2 plateau  $\eta_{\leq 2}^{\star, \text{conn}}(U) \rightarrow 2/3$  at  $N = 8$  in fig. 3 is therefore a finite-size artifact: the body-2 numerator and denominator are both  $O(1)$  and their ratio is finite, but the QFI denominator  $F_Q \sim (\ln N)^{3/2}$  of (57) kills  $\eta_{\leq 2}^{\star, \text{conn}}$  logarithmically as  $N \rightarrow \infty$ .

*The body order required to capture  $F_Q$  must grow with  $N$ .* The Wiener combination  $O_{\leq k}^* = \sum_{r=1}^{k-1} c_r \mathcal{O}_\pi^{XY,r} + \dots$  has variance

$$\text{Var}(O_{\leq k}^*) \sim \sum_{r=1}^{k-1} |c_r|^2 \left[ \text{Var}(\mathcal{O}_\pi^{XY,r}) \right] \sim \sum_{r=1}^{k-1} \frac{1}{r} (\ln r)^{1/2}, \quad (48)$$

where the  $|c_r|^2 \sim (\ln r)^{1/2}/r$  scaling follows from the Wiener formula together with the WZW spin two-point function (43). Setting  $\text{Var}(O_{\leq k}^*) \stackrel{!}{=} F_Q/4 \sim (\ln N)^{3/2}$  gives

$$k_\star(N) \sim N, \quad \eta_{\leq k}^{\star, \text{conn}}(U) \rightarrow 1 \text{ requires } k \propto N, \quad (49)$$

i.e. *no fixed-body operator family saturates  $F_Q$  in the thermodynamic limit at the WZW point*; the matched-filter support must scale extensively. The body-4 closed form (36) captures the leading  $1/r^2$  tails inside a 4-window and bumps  $\eta_{\leq 4}^{\star, \text{conn}}(U)$  from  $2/3$  to  $1$  at  $N \leq 8$ , but at  $N = 16, 32, \dots$  the same body-4 matched filter has  $\eta_{\leq 4}^{\star, \text{conn}}(U) \sim (\ln N)^{-3/2}$  in the same way that  $\eta_{\leq 2}^{\star, \text{conn}}$  does. The only operator family whose variance genuinely tracks  $F_Q$  in the thermodynamic limit is the SLD itself (40), whose support is the full chain and whose body-content (41) extends to all  $r, m, p \leq N$ .

**General  $k$ : window block-Lanczos.** For a general window  $W$  of size  $\leq k$  the optimal operator is built from the block-Krylov vectors of  $|\delta G\rangle$ ,

$$|\xi_n^{(W)}\rangle \equiv (\mathcal{P}_W H)^n |\delta G\rangle_W, \quad ||\delta G\rangle_W \equiv P_W |\delta G\rangle, \quad (50)$$

where  $\mathcal{P}_W$  projects onto operators supported on  $W$  and  $H$  is the chain Hamiltonian. The first  $\text{rank}(P_W |\delta G\rangle)$  Krylov vectors span the symplectic projection  $P_{\mathcal{V}_{\leq k}^{\text{conn}}|\psi_0\rangle} |\delta G\rangle$ . The matched filter is

$$O_{\leq k}^{\star} = \sum_W \sum_n [(\Sigma^{(W)})^+ \mathbf{s}^{(W)}]_n O_n^{(W)}, \quad \eta_{\leq k}^{\star, \text{conn}} = \frac{1}{F_Q} \sum_W (\mathbf{s}^{(W)})^\top (\Sigma^{(W)})^+ \mathbf{s}^{(W)}, \quad (51)$$

with  $O_n^{(W)}$  the unique Hermitian operator on  $W$  whose action on  $|\psi_0\rangle$  is  $|\xi_n^{(W)}\rangle$  (constructed by the Gram-Schmidt-orthogonalised Krylov basis on the operator space of  $W$ ). The hierarchy (51) interpolates between  $k = 2$  (only the leading Krylov vector, recovering (33)) and  $k = N$  (the single window exhausts  $\mathcal{H}$ , recovering (31)).

**Compact symmetry-resolved summary.** In summary, the  $J_1$ - $J_2$ - $J_3$  matched filter at the three reference points has the closed forms

$$(D) : O_{\leq 2}^{\star} \propto \sum_{a \in \text{dimer}} (S_{2a-1}^x S_{2a}^x + S_{2a-1}^y S_{2a}^y), \quad \eta_{\leq 2}^{\star, \text{conn}}(D) = 1, \quad (52)$$

$$(C) : O_{\leq 4}^{\star} = \sum_{b \in \text{cluster}} \frac{1}{2} \hat{L}_{B_b}, \quad \eta_{\leq 4}^{\star, \text{conn}}(C) = 1, \quad (53)$$

$$(U) : O_{\leq 4}^{\star} = \frac{1}{2} \hat{L} \quad ((31) \text{ at } k = N = 8, \text{ coincides with } \mathcal{V}_{\leq 4}^{\text{conn}} \text{ here}), \quad \eta_{\leq 4}^{\star, \text{conn}}(U) = 1. \quad (54)$$

At the WZW point only the leading Krylov vector  $\mathcal{O}_\pi^{XY}$  saturates a fraction  $2/3$  of  $F_Q$ ; the remaining  $1/3$  is carried by the higher-order Krylov vectors  $|\xi_{n \geq 1}^{(W)}\rangle$  inside 4-windows (the body-4  $\mathcal{Q}_\pi$  contribution) and is fully captured by  $O_{\leq 4}^{\star}$  of (36); at  $k = 8$  the same state is exhausted by  $\frac{1}{2} \hat{L}$  of (40).

## 7 Discussion

The geometric extraction of  $F_Q$  via the error-propagation identity (1) is mathematically forced: the SLD is the unique Hermitian direction along which the symmetric generator of  $\rho_\theta$  acts on  $\rho$ , and  $\eta[O] \in [0, 1]$  is the squared cosine of the Hilbert-space angle between  $\tilde{O} |\psi_0\rangle$  and the QFI response  $|\delta G\rangle$ . Restricting the optimisation to a real Hermitian subspace turns this identity into a structural diagnostic.

**The body-order question.** Our central no-go result is theorem 1: the magnitude of  $F_Q$  does not constrain the body order of the cheapest  $F_{\text{EP}}$ -saturating witness. Cluster- product GHZ states realise every value of  $F_Q$  with arbitrary body-order content of the SLD, and the body-order dead zone extends through every order strictly below the cluster size. No general theorem of the form “ $F_Q$  small  $\Rightarrow$  low-body witness suffices” is available, and any practical extraction protocol must explicitly handle the high-body case.



**Strengthening Hyllus–Tóth.** This obstruction is equivalent (by contrapositive) to the local-window matched-filter witness of corollary 2: a violation  $\eta_{\leq k}^{\star, \text{loc}} < 1$  certifies entanglement depth  $> k$  for any one-body  $G$ , regardless of the value of  $F_Q/N$ . This is a genuine strengthening of Hyllus–Tóth, as the cluster-product GHZ example shows that low  $F_Q/N$  is consistent with arbitrarily large dead-zone edge. Conversely, the WZW Heisenberg ground state shows the criterion is sharp on critical states, where the edge tracks the full window size.

**Choice of subspace.** The strict inclusion  $\mathcal{V}_{\leq k}^{\text{loc}} \subseteq \mathcal{V}_{\leq k}^{\text{conn}} \subseteq \mathcal{V}_{\leq k}^{\text{body}}$  orders the three candidate matched-filter restrictions by stringency. The inclusive local-window  $\mathcal{V}_{\leq k}^{\text{loc}}$  is uniquely characterised among them by its one-to-one correspondence with  $k$ -producibility (theorem 2 and corollary 2), and is therefore the correct mathematical gauge for entanglement depth. The on-shell variants ( $\mathcal{V}_k^{\text{loc},=}$  etc.) are useful as scale diagnostics but do not by themselves witness depth. (We write  $\mathcal{V}_k^{\text{loc},=}$  etc. for the on-shell versions.)

**Outlook.** The matched-filter dead-zone edge  $k^\star(|\psi_0\rangle) \equiv \min\{k : \eta_{\leq k}^{\star, \text{loc}} = 1\}$  is an integer-valued order parameter that interpolates continuously between Hyllus–Tóth’s coarse partite-entanglement bound and the fine-grained Schmidt-rank structure of  $|\psi_0\rangle$ . Three natural extensions follow: (i) finite-temperature  $\rho_\beta$  via the spectral SLD (6); (ii) higher-dimensional lattices, where  $\mathcal{V}_{\leq k}^{\text{loc}}$  becomes the union over local neighbourhoods of size  $\leq k$ ; (iii) experimental estimation of  $\eta_{\leq k}^{\star, \text{loc}}$  from Robertson-ratio sample means in the spirit of single-correlator witnesses, which we leave for separate work.

## A Derivation of the matched-filter formula (12)

With  $s_a = -i\langle [O_a, G] \rangle_\rho$  and  $\Sigma_{ab} = \frac{1}{2}\langle \{\delta O_a, \delta O_b\} \rangle_\rho$  as in (11), the Rayleigh quotient  $R(\mathbf{w}) = (\mathbf{w}^\top \mathbf{s})^2 / (\mathbf{w}^\top \Sigma \mathbf{w})$  is maximised by the generalised eigenproblem  $\mathbf{s} \mathbf{s}^\top \mathbf{w} = R \Sigma \mathbf{w}$ , whose only non-trivial eigenvector is  $\mathbf{w}^\star = \Sigma^+ \mathbf{s}$  with eigenvalue  $R^\star = \mathbf{s}^\top \Sigma^+ \mathbf{s}$ . Dividing by  $F_Q$  gives the first equality in (12). For the second, on a pure state we have  $s_a = -i\langle [O_a, G] \rangle = -2 \text{Im} \langle \tilde{O}_a \psi_0 | \delta G \rangle$  and  $\Sigma_{ab} = \text{Re} \langle \tilde{O}_a \psi_0 | \tilde{O}_b \psi_0 \rangle$ , so writing the real-form vectors  $\phi_a^{\text{real}}$  and  $t$  as in the main text,  $\mathbf{s}^\top \Sigma^+ \mathbf{s} / F_Q = \|P_{\text{col}(\Phi_{\text{real}})} t\|^2 / \|t\|^2$ , which is the squared symplectic projection onto  $\mathcal{V}|\psi_0\rangle$ .

## B Detailed derivations of the closed-form matched filters

The closed forms of the matched filter  $O_{\leq k}^\star$  in terms of  $S^\alpha$  are stated in section 6.5 of the main text. We collect here the full derivations and the symmetry restrictions that drive them.

**State-image projection (any  $\mathcal{V}$ ).** The matched filter is the unique  $O^\star \in \mathcal{V}$  whose centred action on  $|\psi_0\rangle$  realises the Hilbert-space projection of  $-2i|\delta G\rangle$  onto  $\tilde{\mathcal{V}}|\psi_0\rangle \equiv \text{span}_{\mathbb{R}}\{\tilde{O}_a|\psi_0\rangle\}$ :

$$\tilde{O}^\star |\psi_0\rangle = -2i P_{\tilde{\mathcal{V}}|\psi_0} |\delta G\rangle, \quad O^\star = \sum_a w_a^\star O_a, \quad \mathbf{w}^\star = \Sigma^+ \mathbf{s}. \quad (55)$$

Off-shell ( $O^\star$  acting on  $|\psi_0\rangle^\perp \setminus \tilde{\mathcal{V}}|\psi_0\rangle$ ) the matched filter is undetermined and may be set to zero with no change in  $\eta^\star$ . This is the unique freedom in  $O^\star$ .

**$k = N$  derivation.** When  $\mathcal{V}$  contains all Hermitian operators on  $\mathcal{H}$ , the projection in (55) is the identity, and the unique minimum-Hilbert–Schmidt-norm representative is the half-SLD  $\frac{1}{2}\hat{L} = i(|\psi_0\rangle\langle\delta G| - |\delta G\rangle\langle\psi_0|)$ , i.e. (31). The HS norm is  $\|\frac{1}{2}\hat{L}\|_{\text{HS}}^2 = \frac{1}{2}F_Q$ .

**$k = 2$  derivation: the staggered XY exchange.** The symmetry quantum numbers of  $|\delta G\rangle$  for  $G = S_\pi^z$  are momentum  $\pi$ ,  $\Delta S_{\text{tot}}^z = 0$ , parity-even, T-even, even body (??); the  $\Delta S^z = 0$  condition is the selection rule that  $G$  is diagonal in  $S^z$  and  $|\psi_0\rangle \in S_{\text{tot}}^z = 0$ . At body two only two real-coefficient Hermitian Pauli strings have  $\Delta S^z = 0$ :  $S_j^z S_{j+1}^z$  (which commutes with  $G$  and contributes nothing to the signal) and the staggered XY exchange  $X_j^{XY} \equiv S_j^x S_{j+1}^x + S_j^y S_{j+1}^y = \frac{1}{2}(S_j^+ S_{j+1}^- + S_j^- S_{j+1}^+)$ . The symmetric chirality  $S^x S^y + S^y S^x = -\frac{i}{2}(S^+ S^+ - S^- S^-)$  is  $\Delta S^z = \pm 2$  and so produces zero signal on the  $S_{\text{tot}}^z = 0$  ground state. A direct commutator computation gives

$$[X_j^{XY}, S_j^z - S_{j+1}^z] = 2i(S_j^y S_{j+1}^x - S_j^x S_{j+1}^y), \quad [X_j^{XY}, S_j^z + S_{j+1}^z] = 0, \quad (56)$$

so  $-i[X_j^{XY}, S_\pi^z] = 2(-1)^j(S_j^y S_{j+1}^x - S_j^x S_{j+1}^y)$  has the same staggered-momentum sector as  $|\delta G\rangle$ . Hence on a translation-invariant ground state the only matched-filter direction in  $\mathcal{V}_{\leq 2}^{\text{conn}}$  is the staggered sum  $\mathcal{O}_\pi^{XY}$  of (32), with the closed form  $\mathcal{O}_{k=2}^* \propto \mathcal{O}_\pi^{XY}$  of (33) and saturation

$$\eta_{\leq 2}^{\star, \text{conn}} = \frac{|[\mathcal{O}_\pi^{XY}, G]|^2}{\text{Var}(\mathcal{O}_\pi^{XY}) F_Q} = \frac{4 |\sum_j (-1)^j \langle S_j^y S_{j+1}^x - S_j^x S_{j+1}^y \rangle|^2}{N \text{Var}(\mathcal{O}_\pi^{XY}) F_Q}. \quad (57)$$

On the dimer-product ground state translation symmetry is broken to dimer translations; the correct matched filter is the sublattice-resolved  $\mathcal{O}_{j \in \text{dimer}}^{XY}$ , which gives  $\eta_{\leq 2}^{\star, \text{conn}}(D) = 1$  as proved in theorem 2.

**$k = 4$  derivation: explicit four-component Wiener problem.** On  $\mathcal{V}_{\leq 4}^{\text{conn}}$  the same  $\Delta S^z = 0$  selection rule eliminates all body-2 strings except the staggered XY exchanges  $\mathcal{O}_\pi^{XY, r}$  at  $r = 1, 2, 3$  defined in (34), plus all body-4 staggered strings of  $\Delta S^z = 0$  Hermitian type, of which the unique translation-symmetry-irreducible representative on a contiguous 4-window is the staggered double-bond hop  $\mathcal{Q}_\pi$  of (35). The Wiener problem therefore collapses to the  $4 \times 4$  closed-form linear system (36); the saturation reads off as (38). On a 4-cluster product state the per-block half-SLD additivity (theorem 2) re-expresses  $(c_1, c_2, c_3, c_4)$  as the per-block SLD coefficients (39). At the WZW point ( $U$ ) the  $\mathcal{Q}_\pi$  component supplies the  $\frac{1}{3}$  of the QFI that the body-2  $\mathcal{O}_\pi^{XY, 1}$  alone misses.

**General  $k$ : block-Krylov spans the response.** The first  $\text{rank}(P_W |\delta G\rangle)$  of the Krylov vectors  $|\xi_n^{(W)}\rangle = (\mathcal{P}_W H)^n |\delta G\rangle_W$  span the symplectic projection  $P_{\mathcal{V}_{\leq k}^{\text{conn}}|\psi_0\rangle} |\delta G\rangle$  on each window; their union over  $W$  spans  $P_{\mathcal{V}_{\leq k}^{\text{conn}}|\psi_0\rangle} |\delta G\rangle$  globally. The Wiener formula on this basis gives (51), which interpolates between  $k = 2$  (single Krylov vector, (33)) and  $k = N$  (single window exhausting  $\mathcal{H}$ , (31)).

## References

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